

Internet of Things (IoT) — CodeAlpha

Internet of Things (IoT) — Internship Overview

This internship program by **CodeAlpha** is designed to provide students with hands-on experience in the **Internet of Things (IoT)**, where physical devices connect and communicate over the internet. Interns will learn about **IoT architecture, sensors, microcontrollers (Arduino/Raspberry Pi), communication protocols (MQTT, HTTP), and cloud integration**.

The program focuses on building **IoT-based applications** for smart homes, healthcare, agriculture, and industrial automation. Through guided mentorship and real-world assignments, interns will gain exposure to **device programming, data collection, and system integration**. By the end of the internship, participants will be able to design and implement **basic IoT projects** and understand the role of IoT in shaping future technologies like **AI, 5G and Smart Cities**.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete file to **Google drive**:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 2 or 3 out of the 4 tasks** listed below (from your domain).
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Internet of Things (IoT) Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: Research Report – IoT in Real Life

Write a **report (800–1000 words)** on IoT applications in one of the following:

- Smart Homes
- Smart Agriculture
- Healthcare Monitoring
- Industrial Automation

Submit in **PDF format with references**.

✓ TASK 2: Sensor-Based Simulation

- Simulate an IoT system using **Tinkercad/Proteus** (no hardware required).
 - Example: LED on/off using a sensor (Temperature/Light/Motion).
 - Submit **simulation screenshots + code explanation**.
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✓ TASK 3: IoT Prototype with Arduino/Raspberry Pi (Optional if hardware available)

Build a simple IoT project, such as:

- Smart door lock system
- Temperature & humidity monitoring system
- Motion detection alarm

Submit **code + circuit diagram + output images**.

✓ TASK 4: Mini Project – IoT Case Study

Prepare a **5–6 page case study** on one of the following:

- IoT security challenges
- IoT & Artificial Intelligence integration
- Future of IoT in Smart Cities

Submit in **PDF format**.



Submission Process

1. Complete **any 2 or 3 tasks**.
 2. Upload all files (PDFs, code files, screenshots, images, reports) into a **Google Drive folder**:
[CodeAlpha_YourName_IoT](#)
 3. Make the folder **shareable (Anyone with the link can view)**.
 4. Paste the folder link in the **Submission Form**.
 5. Optional: Post your IoT project demo on **LinkedIn/YouTube** and tag **@CodeAlpha**.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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